1.11.5 Difference between Overhead line and Underground cables

Sl. No.	Overhead Lines	Underground Cables
1	To increase the load, conductors can be included easily. To increase the working voltage, it is enough to change the insulator.	Change the new cable for two reasons like to increase the load and the conductors cannot be included easily.
2	If need, load can be easily increased.	Load cannot be increased, otherwise cable will get damage.
3	More space is required to install.	Less space is required to install.
4	Maintenance is easy.	Maintenance is complicated.
5	Easy to find the fault.	Difficult to find the fault.
6	Power factor loss is high.	Power factor loss is low.
7	It is suitable for long transmission	It is suitable for short transmission
8	It can be affected by lightning.	It cannot be affected by lightning.
9	Installation cost is low.	Installation cost is high.
10	It is cheap	It is costly.

Points to remember

- The power supply from power plants to the consumers is called power transmission.
- A large amount of alternating current is produced at power stations.
- The specific resistance of copper is less than aluminum.
- Aluminum Conductor with Steel Reinforced is extensively used for power transmission.
- Wooden poles are not currently used.
- The Lattice steel tower is suitable for the distance between 100 meters and 300 meters.
- Bluish green sparking around the conductor is called corona.
- Types of guarding
  i. Cradle guarding ii. Cage guarding
- The insulator used in the stay wire is called stay insulator.
- When electrical power is unable to transmit by overhead lines area underground cables are used.
- The underground cable is not affected by lightning

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